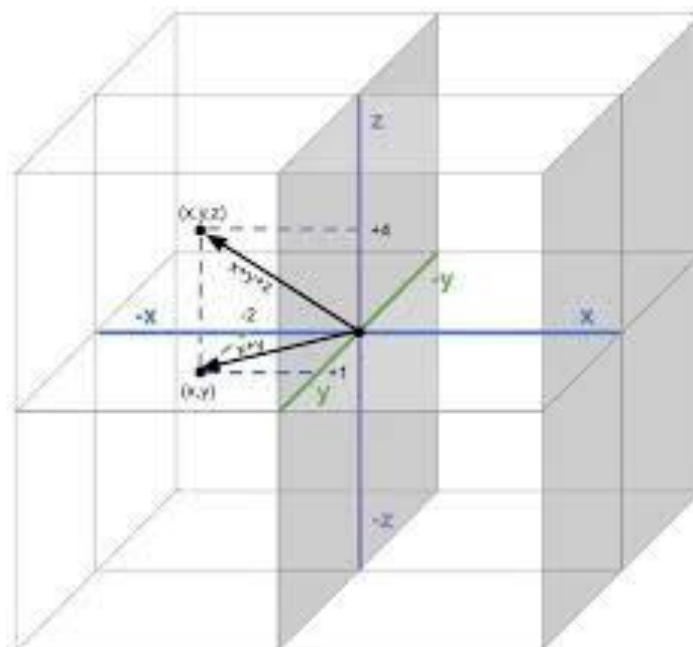


Linear Algebra

Math 1057

Instructor

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Lecture 12

Properties of the Determinant

12.1 ERO and Determinants

Recall that for a matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ we defined

$$\det \mathbf{A} = a_{j1}C_{j1} + a_{j2}C_{j2} + \cdots + a_{jn}C_{jn}$$

where the number $C_{jk} = (-1)^{j+k} \det \mathbf{A}_{jk}$ is called the (j, k) -cofactor of $\mathbf{A}$ and

$$\mathbf{a}_j = [a_{j1} \quad a_{j2} \quad \cdots \quad a_{jn}]$$

denotes the j th row of $\mathbf{A}$. Notice that

$$\det \mathbf{A} = [a_{j1} \quad a_{j2} \quad \cdots \quad a_{jn}] \begin{bmatrix} C_{j1} \\ C_{j2} \\ \vdots \\ C_{jn} \end{bmatrix}.$$

If we let $\mathbf{c}_j = [C_{j1} \quad C_{j2} \quad \cdots \quad C_{jn}]$ then

$$\det \mathbf{A} = \mathbf{a}_j \cdot \mathbf{c}_j^T.$$

In this lecture, we will establish properties of the determinant under elementary row operations and some consequences. The following theorem describes how the determinant behaves under elementary row operations of Type 1.

Theorem 12.1: Suppose that $\mathbf{A} \in \mathbb{R}^{n \times n}$ and let $\mathbf{B}$ be the matrix obtained by interchanging two rows of $\mathbf{A}$. Then $\det \mathbf{B} = -\det \mathbf{A}$.

Proof. Consider the 2×2 case. Let $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and let $\mathbf{B} = \begin{bmatrix} a_{21} & a_{22} \\ a_{11} & a_{12} \end{bmatrix}$. Then

$$\det \mathbf{B} = a_{12}a_{21} - a_{11}a_{22} = -(a_{11}a_{22} - a_{12}a_{21}) = -\det \mathbf{A}.$$

The general case is proved by induction. □

This theorem leads to the following corollary.

Corollary 12.2: If $\mathbf{A} \in \mathbb{R}^{n \times n}$ has two rows (or two columns) that are equal then $\det(\mathbf{A}) = 0$.

Proof. Suppose that $\mathbf{A}$ has rows j and k that are equal. Let $\mathbf{B}$ be the matrix obtained by interchanging rows j and k . Then by the previous theorem $\det \mathbf{B} = -\det \mathbf{A}$. But clearly $\mathbf{B} = \mathbf{A}$, and therefore $\det \mathbf{B} = \det \mathbf{A}$. Therefore, $\det(\mathbf{A}) = -\det(\mathbf{A})$ and thus $\det \mathbf{A} = 0$. $\square$

Now we consider how the determinant behaves under elementary row operations of Type 2.

Theorem 12.3: Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ and let $\mathbf{B}$ be the matrix obtained by multiplying a row of $\mathbf{A}$ by β . Then $\det \mathbf{B} = \beta \det \mathbf{A}$.

Proof. Suppose that $\mathbf{B}$ is obtained from $\mathbf{A}$ by multiplying the j th row by β . The rows of $\mathbf{A}$ and $\mathbf{B}$ different from j are equal, and therefore

$$\mathbf{B}_{jk} = \mathbf{A}_{jk}, \quad \text{for } k = 1, 2, \dots, n.$$

In particular, the (j, k) cofactors of $\mathbf{A}$ and $\mathbf{B}$ are equal. The j th row of $\mathbf{B}$ is $\beta \mathbf{a}_j$. Then, expanding $\det \mathbf{B}$ along the j th row:

$$\begin{aligned} \det \mathbf{B} &= (\beta \mathbf{a}_j) \cdot \mathbf{c}_j^T \\ &= \beta (\mathbf{a}_j \cdot \mathbf{c}_j^T) \\ &= \beta \det \mathbf{A}. \end{aligned}$$

$\square$

Lastly we consider Type 3 elementary row operations.

Theorem 12.4: Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ and let $\mathbf{B}$ be the matrix obtained from $\mathbf{A}$ by adding β times the k th row to the j th row. Then $\det \mathbf{B} = \det \mathbf{A}$.

Proof. For any matrix $\mathbf{A}$ and any row vector $\mathbf{r} = [r_1 \ r_2 \ \dots \ r_n]$ the expression

$$\mathbf{r} \cdot \mathbf{c}_j^T = r_1 C_{j1} + r_2 C_{j2} + \dots + r_n C_{jn}$$

is the determinant of the matrix obtained from $\mathbf{A}$ by replacing the j th row with the row $\mathbf{r}$. Therefore, if $k \neq j$ then

$$\mathbf{a}_k \cdot \mathbf{c}_j^T = 0$$

since then rows k and j are equal. The j th row of $\mathbf{B}$ is $\mathbf{b}_j = \mathbf{a}_j + \beta \mathbf{a}_k$. Therefore, expanding $\det \mathbf{B}$ along the j th row:

$$\begin{aligned}\det \mathbf{B} &= (\mathbf{a}_j + \beta \mathbf{a}_k) \cdot \mathbf{c}_j^T \\ &= \mathbf{a}_j \cdot \mathbf{c}_j^T + \beta (\mathbf{a}_k \cdot \mathbf{c}_j^T) \\ &= \det \mathbf{A}.\end{aligned}$$

□

Example 12.5. Suppose that $\mathbf{A}$ is a 4×4 matrix and suppose that $\det \mathbf{A} = 11$. If $\mathbf{B}$ is obtained from $\mathbf{A}$ by interchanging rows 2 and 4, what is $\det \mathbf{B}$?

Solution. Interchanging (or swapping) rows changes the sign of the determinant. Therefore,

$$\det \mathbf{B} = -11.$$

□

Example 12.6. Suppose that $\mathbf{A}$ is a 4×4 matrix and suppose that $\det \mathbf{A} = 11$. Let $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4$ denote the rows of $\mathbf{A}$. If $\mathbf{B}$ is obtained from $\mathbf{A}$ by replacing row $\mathbf{a}_3$ by $3\mathbf{a}_1 + \mathbf{a}_3$, what is $\det \mathbf{B}$?

Solution. This is a Type 3 elementary row operation, which preserves the value of the determinant. Therefore,

$$\det \mathbf{B} = 11.$$

□

Example 12.7. Suppose that $\mathbf{A}$ is a 4×4 matrix and suppose that $\det \mathbf{A} = 11$. Let $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4$ denote the rows of $\mathbf{A}$. If $\mathbf{B}$ is obtained from $\mathbf{A}$ by replacing row $\mathbf{a}_3$ by $3\mathbf{a}_1 + 7\mathbf{a}_3$, what is $\det \mathbf{B}$?

Solution. This is not quite a Type 3 elementary row operation because $\mathbf{a}_3$ is multiplied by 7. The third row of $\mathbf{B}$ is $\mathbf{b}_3 = 3\mathbf{a}_1 + 7\mathbf{a}_3$. Therefore, expanding $\det \mathbf{B}$ along the third row

$$\begin{aligned}\det \mathbf{B} &= (3\mathbf{a}_1 + 7\mathbf{a}_3) \cdot \mathbf{c}_3^T \\ &= 3\mathbf{a}_1 \cdot \mathbf{c}_3^T + 7\mathbf{a}_3 \cdot \mathbf{c}_3^T \\ &= 7(\mathbf{a}_3 \cdot \mathbf{c}_3^T) \\ &= 7 \det \mathbf{A} \\ &= 77\end{aligned}$$

□

Example 12.8. Suppose that $\mathbf{A}$ is a 4×4 matrix and suppose that $\det \mathbf{A} = 11$. Let $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4$ denote the rows of $\mathbf{A}$. If $\mathbf{B}$ is obtained from $\mathbf{A}$ by replacing row $\mathbf{a}_3$ by $4\mathbf{a}_1 + 5\mathbf{a}_2$, what is $\det \mathbf{B}$?

Solution. Again, this is not a Type 3 elementary row operation. The third row of $\mathbf{B}$ is $\mathbf{b}_3 = 4\mathbf{a}_1 + 5\mathbf{a}_2$. Therefore, expanding $\det \mathbf{B}$ along the third row

$$\begin{aligned} \det \mathbf{B} &= (4\mathbf{a}_1 + 5\mathbf{a}_2) \cdot \mathbf{c}_3^T \\ &= 4\mathbf{a}_1 \cdot \mathbf{c}_3^T + 5\mathbf{a}_2 \cdot \mathbf{c}_3^T \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

□

12.2 Determinants and Invertibility of Matrices

The following theorem characterizes invertibility of matrices with the determinant.

Theorem 12.9: A square matrix $\mathbf{A}$ is invertible if and only if $\det \mathbf{A} \neq 0$.

Proof. Beginning with the matrix $\mathbf{A}$, perform elementary row operations and generate a sequence of matrices $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_p$ such that $\mathbf{A}_p$ is in row echelon form and thus triangular:

$$\mathbf{A} \sim \mathbf{A}_1 \sim \mathbf{A}_2 \sim \dots \sim \mathbf{A}_p.$$

Thus, matrix $\mathbf{A}_i$ is obtained from $\mathbf{A}_{i-1}$ by performing one of the elementary row operations. From Theorems 12.1, 12.3, 12.4, if $\det \mathbf{A}_{i-1} \neq 0$ then $\det \mathbf{A}_i \neq 0$. In particular, $\det \mathbf{A} = 0$ if and only if $\det \mathbf{A}_p = 0$. Now, $\mathbf{A}_p$ is triangular and therefore its determinant is the product of its diagonal entries. If all the diagonal entries are non-zero then $\det \mathbf{A} = \det \mathbf{A}_p \neq 0$. In this case, $\mathbf{A}$ is invertible because there are $r = n$ leading entries in $\mathbf{A}_p$. If a diagonal entry of $\mathbf{A}_p$ is zero then $\det \mathbf{A} = \det \mathbf{A}_p = 0$. In this case, $\mathbf{A}$ is not invertible because there are $r < n$ leading entries in $\mathbf{A}_p$. Therefore, $\mathbf{A}$ is invertible if and only if $\det \mathbf{A} \neq 0$. □

12.3 Properties of the Determinant

The following theorem characterizes how the determinant behaves under scalar multiplication of matrices.

Theorem 12.10: Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ and let $\mathbf{B} = \beta \mathbf{A}$, that is, $\mathbf{B}$ is obtained by multiplying every entry of $\mathbf{A}$ by β . Then $\det \mathbf{B} = \beta^n \det \mathbf{A}$.

Proof. Consider the 2×2 case:

$$\begin{aligned}
 \det(\beta \mathbf{A}) &= \begin{vmatrix} \beta a_{11} & \beta a_{12} \\ \beta a_{12} & \beta a_{22} \end{vmatrix} \\
 &= \beta a_{11} \cdot \beta a_{22} - \beta a_{12} \cdot \beta a_{21} \\
 &= \beta^2 (a_{11}a_{22} - a_{12}a_{21}) \\
 &= \beta^2 \det \mathbf{A}.
 \end{aligned}$$

Thus, the statement holds for 2×2 matrices. Consider a 3×3 matrix $\mathbf{A}$. Then

$$\begin{aligned}
 \det(\beta \mathbf{A}) &= \beta a_{11} |\beta \mathbf{A}_{11}| - \beta a_{12} |\beta \mathbf{A}_{12}| + \beta a_{13} |\beta \mathbf{A}_{13}| \\
 &= \beta a_{11} \beta^2 |\mathbf{A}_{11}| - \beta a_{12} \beta^2 |\mathbf{A}_{12}| + \beta a_{13} \beta^2 |\mathbf{A}_{13}| \\
 &= \beta^3 (a_{11} |\mathbf{A}_{11}| - a_{12} |\mathbf{A}_{12}| + a_{13} |\mathbf{A}_{13}|) \\
 &= \beta^3 \det \mathbf{A}.
 \end{aligned}$$

The general case can be treated using mathematical induction on n . □

Example 12.11. Suppose that $\mathbf{A}$ is a 4×4 matrix and suppose that $\det \mathbf{A} = 11$. What is $\det(3\mathbf{A})$?

Solution. We have

$$\begin{aligned}
 \det(3\mathbf{A}) &= 3^4 \det \mathbf{A} \\
 &= 81 \cdot 11 \\
 &= 891
 \end{aligned}$$

□

The following theorem characterizes how the determinant behaves under matrix multiplication.

Theorem 12.12: Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. Then

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}).$$

Corollary 12.13: For any square matrix $\det(\mathbf{A}^k) = (\det \mathbf{A})^k$.

Corollary 12.14: If $\mathbf{A}$ is invertible then

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det \mathbf{A}}.$$

Proof. From $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_n$ we have that $\det(\mathbf{A}\mathbf{A}^{-1}) = 1$. But also

$$\det(\mathbf{A}\mathbf{A}^{-1}) = \det(\mathbf{A}) \det(\mathbf{A}^{-1}).$$

Therefore

$$\det(\mathbf{A}) \det(\mathbf{A}^{-1}) = 1$$

or equivalently

$$\det \mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}.$$

□

Example 12.15. Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be $n \times n$ matrices. Suppose that $\det \mathbf{A} = 3$, $\det \mathbf{B} = 0$, and $\det \mathbf{C} = 7$.

- (i) Is $\mathbf{AC}$ invertible?
- (ii) Is $\mathbf{AB}$ invertible?
- (iii) Is $\mathbf{ACB}$ invertible?

Solution. (i): We have $\det(\mathbf{AC}) = \det \mathbf{A} \det \mathbf{C} = 3 \cdot 7 = 21$. Thus, $\mathbf{AC}$ is invertible.

(ii): We have $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B} = 3 \cdot 0 = 0$. Thus, $\mathbf{AB}$ is not invertible.

(iii): We have $\det(\mathbf{ACB}) = \det \mathbf{A} \det \mathbf{C} \det \mathbf{B} = 3 \cdot 7 \cdot 0 = 0$. Thus, $\mathbf{ACB}$ is not invertible. □

After this lecture you should know the following:

- how the determinant behaves under elementary row operations
- that $\mathbf{A}$ is invertible if and only if $\det \mathbf{A} \neq 0$
- that $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$